F79SP Session 19: Poisson Processes

1 Bernoulli process

Recall that a Bernoulli Process $\{X_n\}_{n\geq 0}$ is a discrete-time stochastic process such that X_n's are i.i.d Bernoulli r.v.'s.

Some basic properties:

- Events in disjoint time slots are independent.
- Memoryless

Some interesting questions:

- Arrival time:
- No of arrivals in n (consecutive) time slots:

2 Poisson Process: first definition

We can think of Poisson Processes as the continuous-time analogue of Bernoulli processes.

In practice, we consider Poisson process as counting process (which count points).

Definition 2.1 (First definition of PP)**.** *Consider the following setup:*

- *Let $Y_1, Y_2, \ldots$ be i.i.d.r.v.'s following an $Exp(\lambda)$ distribution.*
- *Let $S_0 = 0$, $S_n = Y_1 + \cdots + Y_n$ for $n = 1, 2, \ldots$.*
- *For $t \geq 0$, let $N(t) = max\{n : S_n \leq t\}$.*

Then the (continuous-time) stochastic process $\{N(t)\}_{t\geq 0}$ is called a (homogeneous) Poisson Process.

Note: Exponential distribution can be seen as continuous version of geometric distribution.

3 Main property

We are often interested with the number of arrivals/points in a time interval of a Poisson Process. This is described by result below. Denote $N(t_1, t_2) = N(t_1) - N(t_2)$.

Theorem 3.1 (Property PP1)**.** *Let $\{N(t)\}_{t\geq 0}$ be a Poisson Process.*

1. *Then for any $0 \leq t_1 < t_2$, $N(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1))$.*
2. *For $0 \leq t_1 < t_2 < \cdots < t_l$, the random variables $N(t_1), N(t_1, t_2), \ldots, N(t_{l-1}, t_l)$ are mutually independent.*

Proof. We will show that First Definition implies Theorem 3.1. □

In words:

1. No of points in an interval follows Poisson distribution with rate proportional to the length of the time interval.
2. No of points in disjoint intervals are independent.

Note: Poisson Theorem shows Poisson distribution as an approximation to binomial distribution.

Corollary 3.2. *Let $\{N(t)\}_{t\geq 0}$ be a Poisson Process. Let $t > 0$ and $n \in \mathbb{Z}^+$ be fixed. Given $N(t) = n$, all n points of the PP on $(0, t)$ behave like mutually independent and uniformly distributed.*

Proof. Outline of proof:

- We can write $N(t) = N(c) + N(c, t)$ for $0 < c < t$.
- Use PP1 to get $N(c) \sim \Pi(\lambda c)$ and $N(c, t) \sim \Pi(\lambda(t - c))$, and they are independent.
- From Property Π2:

$$\begin{aligned} P(N(c) = j | N(t) = n) &= P(N(c) = j | N(t) = n) = P(N(c) = j | N(c) + N(c, t) = n) \\ &= \binom{n}{j} (c/t)^j (1 - c/t)^{n-j} \end{aligned}$$

- Hence, $N(c)|N(t) = n \sim B(n, \frac{c}{t})$. Use Property U3 with $a = 0$ and $b = t$, we have the points location's $X_1, X_2, \ldots, X_n$ are mutually independent and uniformly distributed in $U(0, t)$.

□

Exercise: (Modified from T6Q7) Calls arrive at a police station as a Poisson process with rate $\lambda = 4$ per minute. With respect to any fixed origin in time, find:

1. The probability that the first call arrives after three minutes.
2. The probability that the number of calls arrive in the first four minutes is less than 2.
3. The probability that the number of calls arrive in the first four minutes is less than 2, given that the number of call arrive between 12th to 15th minutes is 3.
4. The probability that the call arrive between the 3rd and 4th minutes, given that the number of calls arrive before the 5th minute is 1.
5. The probability that two of the calls arrive between the 3rd and 4th minutes, given that the number of calls arrive before the 5th minute is 3.
6. The probability that the twelfth subsequent call arrives before two minutes have elapsed.
7. The probability that the twelfth subsequent call arrives after three minutes and before five minutes have elapsed.